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M. Light, J. Sinur

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Sizing Up the SDLC

Clients often ask how much effort the phases of AD should take relative to each other. General guidelines must be adapted to the realities of different project types and development styles.

Core Topics
Applications Development: Applications Development Methodology; Applications Development Project Management

Key Issues
What project management best practices will assist AD organizations in maximizing return on investment for their AD projects while reducing the potential for cost overruns, late delivery and scope creep?

Which processes, products and vendors will provide effective methodology solutions in support of application delivery?

How much effort should the phases of AD take relative to one another?

Figure 1 presents analysis of the systems development life cycle (SDLC) phases' relative effort and is a composite of client feedback, experience-based observation and analysis, GartnerMeasurement data and other industry data and trend analysis. Figure 1 and Figure 2 give percentages for average proportions of each phase in both new development and maintenance. Proportions, however, will vary by type of project (e.g., "traditional" large systems, RAD, package integration, or client/server), sometimes widely (e.g., Web site development, where maintenance of content may equal 500 percent of development costs within the first year).

Figure 1
Development Effort for Initial Application Release

	Traditional	C/S
Requirements analysis	15	15
Logical design	5	10
Physical design	15	15
Coding	30	20
Unit testing	15	15
System	10	15
Installation	10	10
Total	100	100

Source: GartnerGroup

New Development
New development typically represents about 30 percent to 50 percent of total life cycle costs; average duration of major projects is 12 to 18 months.

Figure 2 notes the split between development costs and maintenance and enhancement costs. A typical application consumes 15 percent to 20 percent of the resources used to develop it for every year in maintenance, so that in the course of five to seven years the application consumes the same amount of resources in maintenance that it consumed in development. The proportion of maintenance costs will vary according to how much business change affects the application. If the application "lives" for 10 years, the amount of resources for maintenance approaches two-thirds of the application's total cost.

Figure 2
Effort Supporting Application After Initial Release

	Traditional	C/S
Bug Fixing	35	45
*coding	*60	*50
*testing	*40	*50
Enhancements	40	30
*Req'ts	*15	*20
*Logical design	*5	*5
*Physical design	*20	*20
*Coding	*35	*30
*Unit testing	*25	*25
Regression tests	25	25
Total	100	100

Source: GartnerGroup

Support
Maintenance and enhancement typically represent about 50 percent to 70 percent of total life cycle costs; "average" duration in support for major systems ranges widely, from a low of approximately 36 months to as long as 10 years or more.

Relatively low average effort for logical design and relatively high rates for bug fixing in maintenance are indicated, not coincidentally. Best practices in AD use effective logical design upfront to improve quality and application flexibility, so that in many cases "investing" more than usual in this phase often yields substantial benefits. Strong logical design is also a significant element of AD reuse, which can contribute to both quality and development productivity. High rates of bug fixing during maintenance — especially for client/server systems — indicate poor processes for assuring quality.

Acronym Key

AD	Applications development
C/S	Client/server
RAD	Rapid applications development
SDLC	Systems development life cycle

Average enhancement rates may be lower than desirable. Enhancement projects are an attractive alternative to very large development projects because of the latter's tendency toward cancellation or cost/schedule overruns of 200 percent or more. Best practices in large scale AD often involve planned, iterative development cycles (see RDS *Research Note* KA-480-071, 27 November 1995, "RAAD: Is It Being Used Today?").